

DEVELOPMENT PLAN AND TIMELINE

The Movie Theatre Website will be developed incrementally. The team will first confirm the requirements and design, then implement the core customer and administrative features, and finally test the complete system. Work will be tracked through the group GitHub repository. Each team member will create and push commits for the work assigned to them.

Task Partitioning

1. Requirements and project documentation

- Review the original Software Requirements Specification.
- Keep the requirements, architecture, UML diagram, and implementation plan consistent as the project develops.
- Maintain the final project documentation and prepare the final submission.

2. Database and domain model

- Create the relational database tables for movies, theatres, seats, showtimes, reservations, orders, tickets, payments, users, and audit logs.
- Implement the domain objects, identifiers, status values, and validation rules described in the class specifications.

3. Customer website

- Implement movie listings, movie details, showtime selection, seat-map display, ticket selection, checkout, order confirmation, and guest order retrieval.
- Clearly display available, selected, temporarily reserved, sold, and inaccessible seats.

4. Administrator website

- Implement administrator authentication and authorization.
- Implement movie, theatre, showtime, ticket-type, price, and order management functions.
- Record important administrative actions in the audit log.

5. Reservation, checkout, and payment processing

- Implement temporary seat holds and expiration handling.
- Prevent two customers from purchasing the same seat for the same showtime.

- Implement order totals, taxes, service fees, payment approval or failure, and order confirmation.
- Ensure that complete payment-card numbers and security codes are never stored.

6. Testing and quality assurance

- Test normal, invalid, and failure conditions for customer and administrator workflows.
- Test showtime conflicts, seat-hold expiration, duplicate-seat prevention, payment failures, guest order retrieval, and administrator permissions.
- Correct defects and verify that the final implementation agrees with the requirements and design specification.

Team Member Responsibilities

Nate Crivelli

- Lead the application architecture and backend/domain implementation.
- Implement or coordinate the database, repositories, domain classes, reservation logic, checkout logic, and payment-service interfaces.
- Review the security rules and verify that transactional seat and order operations are handled correctly.

Matthew Chung

- Lead project documentation and customer-facing functionality.
- Maintain the requirements/design documentation, UML updates, development plan, and final report sections.
- Implement or coordinate the customer movie listings, movie details, showtime selection, seat-map display, checkout pages, and order retrieval.
- Assist with integration testing, proofreading, defect tracking, and final submission preparation.

Both team members

- Review each other's work before merging changes.
- Use descriptive GitHub commit messages and push at least one commit each.
- Participate in integration testing and approve the final project version.

Development Timeline

Phase 1 - Requirements and design review

Activities: Review the SRS, confirm the architecture, verify the UML class diagram, and finalize the development plan.

Deliverable: Approved requirements and software design specification.

Phase 2 - Database and application foundation

Activities: Create the database schema, repositories, domain classes, authentication foundation, and project configuration.

Deliverable: Working data and application foundation.

Phase 3 - Customer functionality

Activities: Implement movie browsing, showtime selection, seat maps, ticket selection, checkout forms, guest retrieval, and customer order history.

Deliverable: Customer purchase workflow.

Phase 4 - Administration and integrations

Activities: Implement administrator functions, reservation expiration, payment-provider integration, email notifications, and audit logging.

Deliverable: Working administrative and external-service features.

Phase 5 - Integration and system testing

Activities: Connect all layers, test complete workflows, test security and concurrency rules, and correct defects.

Deliverable: Tested integrated system.

Phase 6 - Final review and submission

Activities: Review the implementation against the SRS, update diagrams and documentation, verify that both members have GitHub commits, and submit the final repository link.

Deliverable: Final project report and repository submission.